CIS 220 Database Concepts

Assignment: Install sample databases

Investigating and reverse engineering existing, well designed databases is a great way to wrap your head around the concepts. We will be exploring two of the classic sample databases.

- Sakila
- Classic Models

The SQL files for these databases are attached to this assignment. I noticed when grading your first assignment that the Windows installation of MySQL actually comes with the Sakila database, but the Mac and Linux installations do not. Quite curious.

As you know, you can see the databases that your instance of MySQL server knows by logging into MySQL and executing

show databases;

If you do this and you see Sakila show up in this list you do not need to install it, but you will probably want to review the installation instructions for Sakila, so that you can install the Classic Models database.

Things to consider:

The Sakila database is much larger in size and data volume than Classic Models. This database comes as text file containing a series of SQL statements. There are two files for Sakil. One that contains the SQL to DEFINE the structure of the database, and an additional file that contains the DATA to put into the database. The installation instructions will make this clear.

The installation instructions for Sakila can be found here:
https://dev.mysql.com/doc/sakila/en/sakila-preface.html

To complete this assignment successfully it is important that you understand the difference between being logged into the MySQL monitor and simply being inside your terminal. The prompt is a good indicator. Notice in the square box in the screen shot below. This is showing the **mysql prompt** after you have logged in. At this prompt you can only execute commands that mysql understands. You cannot execute normal OS commands like **cd, dir, ls . . . etc.**

```
❯ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.21 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Above, I am logged into MySQL and the prompt illustrates that.
In the next screenshot, I have exited the MySQL Monitor and am back at my command prompt.

```
❯ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.0.21 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> quit;
Bye
 OSX | ~ ──────────────────────────── took 20s | with Admin@KWhitener-MBP | at 16:20:29
❯
```

Notice how the prompt changes depending on where you are and what you are running. Your prompt will look different from mine, but spend some time logging in and out to get a feel for what is going on.

When you have completed the installation of Sakila and Classic Models I'd like you to take the following screen shots as your submission. **Perform these actions for both Sakila and Classic Models.** I included screen shots of my system for reference.

Task	*MySQL Command*
Show all the databases you are monitoring	**mysql> show databases;**
Select the sakila database	**mysql> use sakila;**
Show all tables in the sakila database	**mysql> show tables;**
Pick a table and show its structure	**mysql> describe actor;**

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| classicmodels      |
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
+--------------------+
6 rows in set (0.02 sec)

mysql> use sakila;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+----------------------------+
| Tables_in_sakila           |
+----------------------------+
| actor                      |
| actor_info                 |
| address                    |
| category                   |
| city                       |
| country                    |
| customer                   |
| customer_list              |
| film                       |
| film_actor                 |
| film_category              |
| film_list                  |
| film_text                  |
| inventory                  |
| language                   |
| nicer_but_slower_film_list |
| payment                    |
| rental                     |
| sales_by_film_category     |
| sales_by_store             |
| staff                      |
| staff_list                 |
| store                      |
+----------------------------+
23 rows in set (0.00 sec)
```

```
mysql> describe actor;
+-------------+------------------+------+-----+-------------------+-----------------------------------------------+
| Field       | Type             | Null | Key | Default           | Extra                                         |
+-------------+------------------+------+-----+-------------------+-----------------------------------------------+
| actor_id    | smallint unsigned | NO   | PRI | NULL              | auto_increment                                |
| first_name  | varchar(45)      | NO   |     | NULL              |                                               |
| last_name   | varchar(45)      | NO   | MUL | NULL              |                                               |
| last_update | timestamp        | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-------------+------------------+------+-----+-------------------+-----------------------------------------------+
4 rows in set (0.01 sec)
```

Connecting to MySQL with MySQL Workbench

It's important to realize that ***MySQL*** and ***MySQL Workbench*** are two separate things. MySQL is the database server that manages the storage, contains the query engine, concurrency manager . . . etc. MySQL Workbench is simply a graphical tool that sits on top of MySQL and allows developers the convenience of not having to execute terminal commands to get things done. It also includes all kinds of performance monitoring and server admin tools.

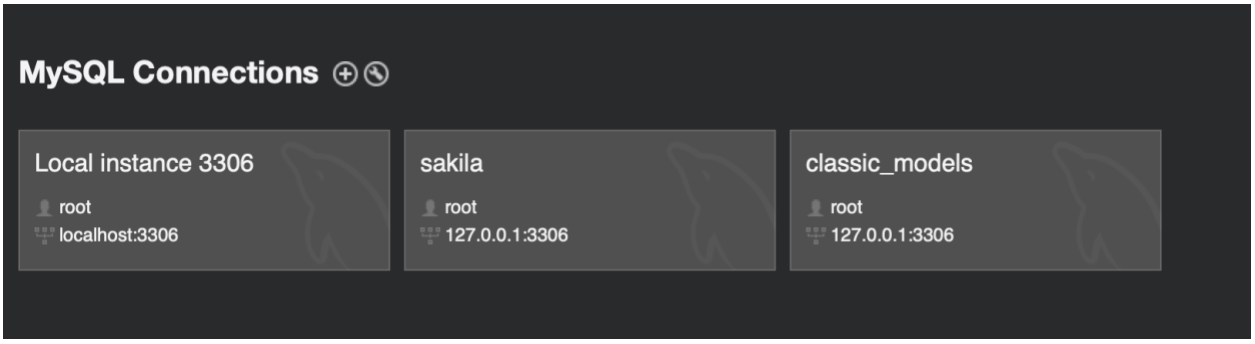
For this portion of the assignment I would like for you to create two new connections. One to the Sakila database and one to the Classic Models database.

The following tutorial will show you how to create connections between the MySQL management tool MySQL Workbench and your local MySQL server.

https://dev.mysql.com/doc/workbench/en/wb-getting-started-tutorial-create-connection.html

When you have completed this portion of the assignment, screen shot the Workbench main window showing the connections. You may have to select this view by clicking on the dolphin DB icon →

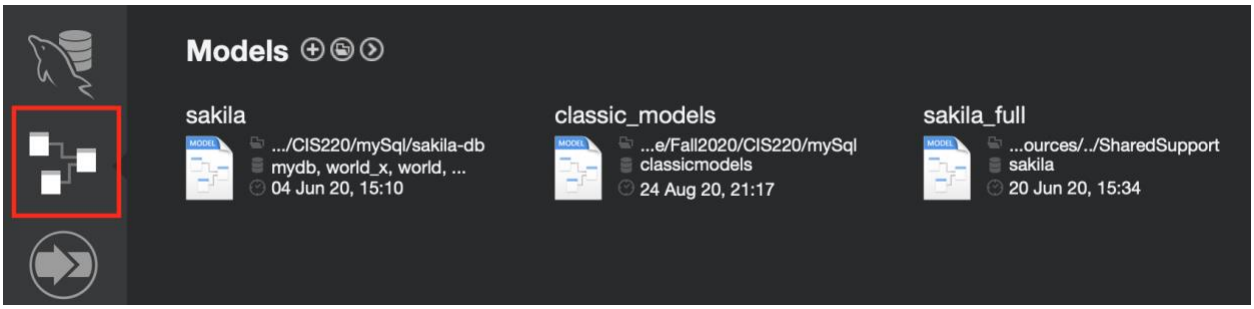




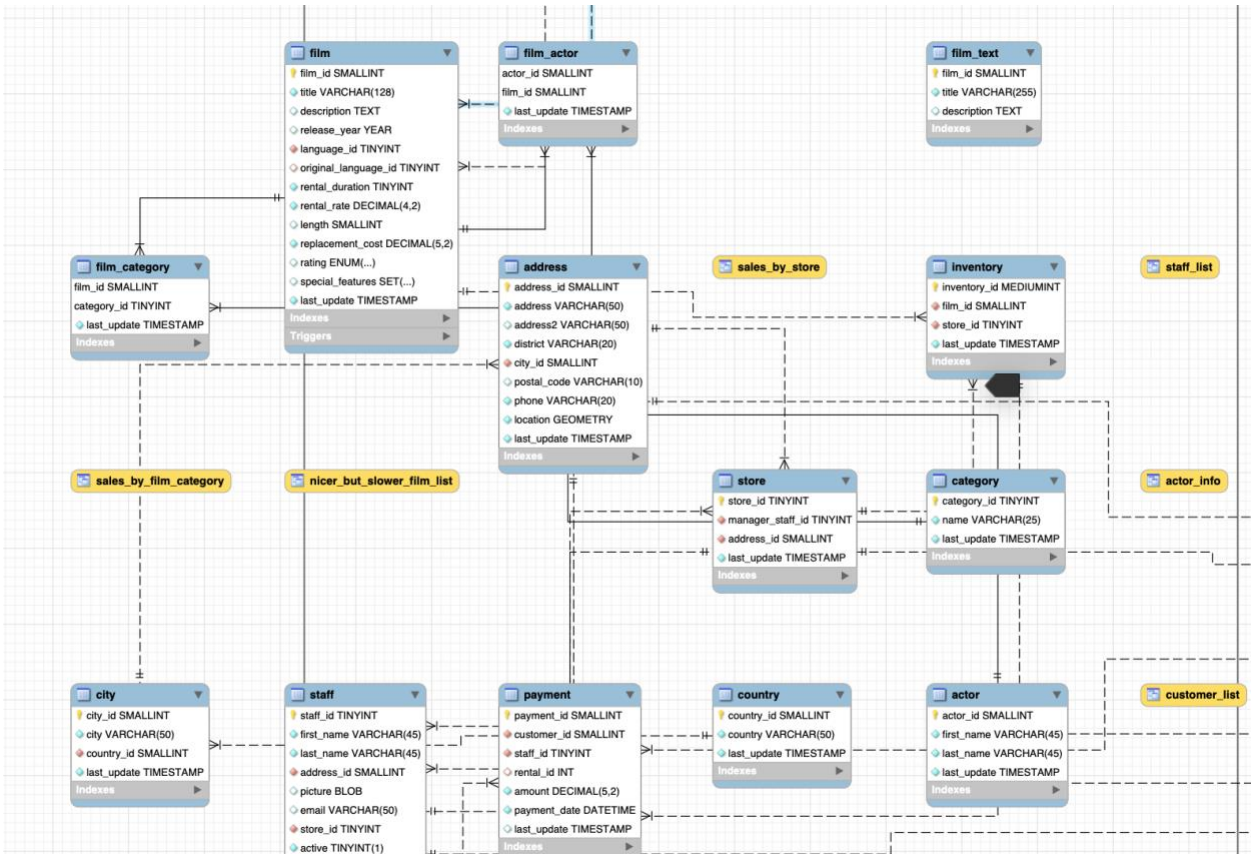
Adding a MWB file to MySQL Workbench. A MWB file is a database document created by MySQL Workbench. This file saves a database design, called an EER (Enhanced Entity-Relationship) model; defines tables, views, routines, privileges, and other constructs for a database.

The Sakila database comes with a design document stored as a MWB file. You should see this as part of the file download. This document allows us to view the structure of a database in a graphical way. This is indispensable when studying and learning about relational databases.

To enter the modeling portion of Workbench, click on the databases schema icon. You can see I already have a couple loaded in and I did a **reverse engineer** of the classic models database. Click the ***folder icon*** to load in the provided MWB file.



When the file is loaded you should be able to see the graphical schema. It may resemble the following



Screenshot a portion of the diagram showing that you were successful, and you are done! Place your screen shots into a document and export as a PDF. Attach the PDF to the submission are of the Blackboard assignment.